

isn't being used it's simply power gated and effectively turned off.

Turbo mode

Talking about multipliers brings around the inevitable topic of overclocking. A decade back, just the mention of overclocking was enough to get people at Intel all riled up. But with Nehalem, Intel is in one sense, encouraging it. The i7 supports a feature called "Turbo Mode" (TM). A processor will be rated to operate within certain constraints — a maximum temperature, current and power consumption. If Turbo Boost is enabled, and if the processor's on-die Power Control Unit, which monitors the thermal condition and power usage of the cores, indicates that it is running within these constraints, one or more cores will start to operate at higher clock speeds, enabling higher performance for the threads they are executing.

When cores are clocked-up in this way, steps of 133.33 MHz are used — the base frequency mentioned earlier. When this increase in clock speed occurs will depend on the conditions inside the machine, most particularly the cooling system. The more efficient the cooling, the more likely that clock speeds will increase. The most likely situation for Turbo Boost to be effective will be when a single- or dual-threaded compute intensive application is running, with little demand on the other cores. In that case, the core running the main thread is very likely to be clocked up, with the other cores not running, thereby reducing power consumption and heat dissipation.

This feature addresses the one criticism of multi-core processors: that not all compute intensive applications have the inherent parallelism that can benefit from multiple cores, and that such processors give no performance improvement for single-threaded applications. With Turbo Boost enabled, such an application can gain extra performance through the increased clock speed of the core on which it is running. This is perhaps the most important feature expressing the scalability that Intel claims to have built into this new architecture.

Cheap thrills

At the outset, it's evident that Intel adopted a very cost-centric approach to designing the i7. It clearly draws on the best points of their previous architectures and with a borrowed trick or two, manages to raise the performance bar considerably. Partly inspired by the Atom design, the i7 has a very low die-area cost. We found Nehalem to be around 20 per cent faster than its predecessor and all this on a new platform that is bound to see optimisations and further performance increments. The LGA 1366 should be a good system to invest in for those looking at a powerful desktop processor; yet the costs related to purchasing a



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new motherboard coupled with the high prices of DDR3 means Nehalem may find few takers in price sensitive markets (such as India). As of now, the X58 platform is the only one available, although NVIDIA has obtained the necessary license from Intel and we could see NVIDIA boards supporting i7 soon. Intel also claims that future eight-core Nehalems will also be designed for socket 1366; this is serious longevity for the platform although whether it keep its promise remains to be seen. The mess of platforms for different processors in the Pentium 4 / Pentium D era is all too fresh in our minds — we hope the 1366-pin socket stays awhile; else early adopters will surely have sufficient grounds for complain.

Parallelism

It seems Intel deliberately chose features that would ensure it excels in the server domain — which has, by far, been its Achilles' Heel over the past two years. In fact, a number of tricks under Nehalem's 45nm hood won't even be noticed unless the i7 is used in multi-socket servers. Most of these tricks involve parallelism and of course more effective caches and prediction systems. Developers for desktops need to get cracking on getting their applications more and more multi-threaded because this is where the i7 will shine and future processors will follow suit. Regardless of intent, the i7 offers a nice performance boost across applications in the desktop space as well. Gaming, 3D rendering, and video encoding are the three biggest beneficiaries. In fact it even delivers improved performance on a core-to-core basis — so Intel has another architectural winner.

Just when we thought it couldn't get much better Intel throws us a few more delightful surprises. Eight-core Nehalems could reach markets by the end of 2009. We've also heard that Intel plans to introduce a new Bloomfield-derivative family named "Lynnfield". Identical to i7; these CPUs sacrifice the QPI for a more traditional DMI and also forego tri-channel DDR3 for a dual-channel setup. These CPUs will utilise a 1156-pin socket and here's the killer — there's an on-package PCI Express controller that supports 16 PCI Express lanes. Yes, the PCIe lanes are on the CPU itself! These 16 lanes can be used as a single x16 connect or two x8 connects. Intel has also discussed "Havendale" which is a dual-core variant with 4 MB of L3 cache. These CPUs will be the first Intel CPUs to have a graphics core integrated as part of the CPU package. It also has the 16 lane PCI Express controller from Lynnfield, but it cannot be run as two X8 connects. Of course, Lynnfield is still a good four months or so from market; while Havendale won't debut till 2010. The only thing better than living in interesting times is the knowledge that the best is yet to come! ■

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